



VIKRAM REDDY ANNEM FULL STACK DEVELOPER

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Professional Summary

- **Full-Stack Engineer** with around 7 years of experience architecting and delivering high-impact web and mobile applications.
- Collaborated with **product managers, designers, QA, and stakeholders in Agile/Scrum** environments using **JIRA, GitLab, and Bitbucket**, conducting **regular standups, sprint planning, and retrospectives**.
- Proponent of modern patterns including **micro-frontends, serverless architecture, performance tuning, and security best practices**.
- Developed performance optimization strategies including **lazy loading patterns, code splitting techniques, and resource prioritization** to achieve optimal page load metrics and user experience scores.
- Pioneered **Adobe Experience Manager (AEM)** implementations including custom component development, template structures, dialog configurations, and content inheritance patterns for large-scale content management solutions
- Back-end architect skilled in **Node/Express, and Java Spring Boot frameworks**, designing microservices with **RESTful APIs, GraphQL/Apollo, WebSocket** used **event-driven architecture** and secure **JWT/OAuth2 authentication**.
- Expert in front-end development using **React, TypeScript, and Next.js (SSR/SSG)**, with **Redux Toolkit, Zustand, Tailwind CSS, Material-UI** and **Storybook**, selective **Angular** expertise for legacy integrations.
- Proficient in using **React JS** for creating reusable components along with **sliders, charts, grids** and **handling events** and **DOM objects**.
- Proficient in **React HOC, React Hooks, React Bootstrap** and **packing tools** like **Vite, Styles** through **material- UI, styled-components**.
- Built **modern Progressive Web Apps** with **service workers, Indexed DB, and offline-first architecture** for improved performance and user experience.
- Database expertise across relational and **NoSQL systems PostgreSQL, MySQL, MongoDB, DynamoDB, and Firestore**.
- Automated testing advocates using **Jest, React Testing Library, Playwright, and Puppeteer** to ensure **code quality** and reliability.
- **CI/CD & DevOps** building pipelines with **GitHub Actions, Jenkins**, and automating builds, tests, and zero-downtime deployments.
- Cloud-native deployments on **AWS (Lambda, API Gateway, EC2, S3, CloudFormation)** and **GCP (Cloud Functions, App Engine)**, with **containerization and orchestration via Docker and Kubernetes**.
- Dedicated mentor and team leader, driving best practices and delivering scalable solutions that exceed business objectives.

Technical Skills

- **Languages:** JavaScript, Java, Python, TypeScript, HTML, CSS3, XML (DOM).
- **Frontend Frameworks:** React JS, Next JS, Angular, Vue JS .
- **State Management & Reactive:** Redux Toolkit, Context API, Octagon, NgRx (Store & Effects), RxJS.
- **UI & Styling:** HTML5, CSS3, SASS/SCSS, Tailwind CSS, Bootstrap, Material-UI, Storybook.
- **Backend Frameworks:** Node JS, Express JS, NestJS, Java Spring Boot.
- **Core Java:** Java 8, Collections, JDBC, Multi-threading, AWT, JSP, Servlets, Ajax, EJB, JMS, Java Beans, Custom Tags, Spring.
- **API Development:** RESTful APIs, GraphQL/Apollo, WebSocket's, JWT, OAuth 2.0

- **ORMs:** Hibernate ORM, Spring Data JPA, Prisma
- **Databases:** MySQL, PostgreSQL, Oracle, MongoDB, DynamoDB, Firebase (Firestore).
- **Cloud Platforms:** AWS (Lambda, EC2, S3, Step Functions, CloudWatch, IAM, Cognito), GCP Firebase Functions.
- **Containerization & Orchestration:** Docker, Kubernetes, EKS.
- **Version Control & CI/CD:** Git, GitLab, Jenkins, GitHub Actions, GitLab CI/CD
- **Testing Libraries:** Jest, React Testing Library, playwright, UAT, Puppeteer, Mocha, Junit.
- **IDE & Development Tools:** VS Code, IntelliJ IDEA, Google Collab, Redux Dev, Chrome DevTools
- **Web Security:** HTTPS, CSP, OWASP, CSRF.
- **AI & Automation:** Prompt Engineering, ChatGPT APIs
- **Performance Tools:** Google Lighthouse, React Profiler, Web Vitals, Page Speed Insights etc.

Professional Experience

Client: Amex, Arizona

Role: Full Stack Developer | React Focused | Aug 2023– Present

Developed **Membership Rewards redemption portal** using **Spring Boot** and **Node.js** with **GraphQL/RESTful APIs**, while **optimizing frontend performance** with **Typescript, Next.js, React**, and **Redux** for seamless user experience.

- **Worked** with cross-functional squad through **Agile Scrum** ceremonies, **architecture reviews**, and best-practice workshop fostering clean-code principles and improving team collaboration.
- Architected a **micro-frontend ecosystem** via **Webpack Module Federation** for the loyalty, fraud-alert, and account-management modules enabling parallel team deployments and **per-feature rollbacks** without impacting other modules.
- Developed the Membership Rewards redemption portal using **Spring Boot Microservices** and **REST/GraphQL** endpoints consumed by React UI.
- Designed and implemented **multi-threaded Java solutions** using **ExecutorService** and **Java 8's Stream API** to optimize parallel processing of high-volume account transactions.
- Improved performance by utilizing **Java 11's HTTP Client API** and **CompletableFuture** for handling asynchronous communication with third-party services.
- Architected and integrated **Gen AI-driven fraud detection pipelines**, connecting **Apache Kafka** transaction streams to ML/AI engines for real-time decisioning in the Authorization system.
- Owned features from **concept to launch** (e.g., Spend Dashboard and Fraud Insights Hub), ensuring accountability across **design, development, and deployment**.
- Refactored **agent servicing components** into reusable libraries with Storybook, improving developer velocity and consistency across multiple portals.
- Collaborated with InfoSec to integrate **3-D Secure** and tokenization services alongside **OAuth 2.0/JWT** authentication, aligning customer-facing endpoints with PCI-DSS standards.
- Developed Card Member Portal by building **Next.js** dynamic dashboards with dynamic imports, **React.Lazy/Suspense**, **Recharts**, and **D3.js** enabling **AI-powered insights**.
- Architected SSR and SSG solutions using **Next.js** for dashboards and public pages, improving initial load performance and SEO for highly dynamic, customer-facing web applications.
- Integrated **React Server Components** alongside server-side APIs to offload rendering logic for offers and receipt flows and reduce client-side JavaScript execution.
- Architected state-management solutions with **Context API** for lightweight global state and **Redux Toolkit** for predictable, scalable client-side stores that interact seamlessly with server APIs.
- Integrated **AI-driven recommendation APIs** to personalize redemption options, boosting customer engagement through contextual insights.
- Used memorization (useMemo, useCallback) to optimize data-intensive widgets, like **real-time LLM chat interfaces** that require minimal re-renders and optimized **Virtual DOM** reconciliation techniques in data-intensive dashboard widgets.

- Cached critical Offer Hub merchant and promotion data in a **Redis** cluster, cutting API response times during peak refresh windows and improving engagement on targeted promotions.
- Implemented **event-driven pipelines** on **Apache Kafka** to feed transaction events into an ML-based fraud detection engine for the Enhanced Authorization real-time authorization system.
- Integrated secure payments (Stripe, PayPal, Razorpay) into conversational payment flows, paving the way for **AI-assisted transaction guidance** within the portal.
- Instituted a three-tier **automated testing** strategy with **Jest** for unit tests, **React Testing Library** for component tests, and **Puppeteer** for end-to-end browser automation.
- Automated end-to-end CI/CD pipelines with **Jenkins** and **GitLab CI**, **Docker** image builds and **Kubernetes** deployments to **AWS EKS**
- Led creation of the company Design System in **Storybook**, curating over 20 interactive **React** components that matched **Figma** prototypes pixel-for-pixel and standardized styling across web portals, mobile apps, and partner integrations.
- Conducted peer **design reviews** and **architectural workshops** that improved scalability and accelerated onboarding of new developers.
- Integrated **Adobe Experience Manager (AEM)** as the enterprise CMS, enabling seamless delivery of promotions, offers, and content across loyalty, fraud-alert, and account-management portals.
- Designed **headless AEM components** consumed by Next.js and React UIs, improving scalability.
- Partnered with content authors to configure **AEM templates, dialogs, and content fragments**, ensuring reusability and alignment with enterprise design system.
- Partnered with **UX** to build annotated component libraries and reusable style tokens in **Figma**, streamlining design-to-development handoffs and ensuring consistent theming across channels
- Employed third-party libraries (**Lodash**, **Formik**, **Framer Motion**) to accelerate development of dynamic forms, animations, and utility functions within payment features.

Client: KDM Engineer PVT LTD | India

Role: Senior React Developer | Full Stack Developer Jan 2020 – Dec 2022

Developed a full-stack **material-testing platform** at **KDM Engineers** using **Spring Boot microservices** and **REST/GraphQL APIs** for backend processing, paired with **React Hook Form**, **React Query**, and **Next.js serverless routes** for frontend UIs, while orchestrating **AWS Lambda**, **Step Functions**, and **DynamoDB** for automated invoicing.

- Extended form handling with **React Query** to fetch live material options from a **Spring Boot catalog API**, ensuring up-to-date selections and **client-side validation**.
- Built a **React Hook Form**-powered order entry UI with **Next.js API middleware** and **authentication**, enabling clients to self-serve material test requests.
- Trained TensorFlow model into the testing workflow to **auto-detect quality issues** and generate AI-authored PDF summaries - a precursor to **Gen AI document summarization** pipelines.
- Designed **React Hook Form + React Query** workflows for client portals that **automated validation and reduced agent rework** time by 25%.
- Built a real-time **material delivery map dashboard** using **Deck.GL** and **React**, plotting test dispatch routes with dynamic tooltips and transition effects to aid logistics teams.
- Built interactive **D3.js dashboards** in Next.js for visualizing material test results and logistics data, including scatter plots, bar charts, and trend lines to highlight anomalies in material quality.
- Collaborated with QA to **document UAT test cases** for React components and serverless backend integrations.
- Designed **custom AEM content fragments** for material testing data, invoices, and logistics updates, consumed by React components via GraphQL endpoints.
- Implemented **Next.js SSR for AEM-powered pages**, optimizing SEO and ensuring test reports were indexed and discoverable for clients.

- Partnered with QA and business users to validate **AEM-React integrations**, ensuring CMS-driven content met compliance and business acceptance criteria.
- Addressed **UAT-reported UI/UX issues**, improving client-facing forms, dashboards, and report layouts.
- Implemented responsive layouts in React with **CSS Grid, Flexbox, and Tailwind CSS**, ensuring **pixel-perfect** experiences from mobile to desktop.
- Implemented a unified cart and checkout flow via **Next.js API routes and Socket.io**, delivering instant order confirmations and reducing latency by 30%.
- Developed interactive **React calculation forms** that posttest parameters to a **Spring Boot microservice** ingesting clickstreams via **Amazon Kinesis** for analytics.
- Integrated **Gen AI document summarization APIs** to auto-generate material test reports and convert raw lab outputs into structured insights for clients.
- Remediated **security vulnerabilities**, deprecated patterns, and compliance gaps across **Java Spring Boot microservices**, ensuring full alignment with corporate secure coding standards.
- Decoupled the order-fulfillment pipeline via **RabbitMQ on Amazon MQ**, ensuring at-least-once delivery guarantees and simplifying microservice scaling.
- Composed **AWS Step Functions** workflows integrating Location Service and **SQS** to dynamically **optimize delivery routes** based on traffic and temperature.
- Leveraged **AWS Amplify CLI** to bootstrap and manage frontend environments, automating deployment pipelines for React/Next.js applications.
- Orchestrated serverless invoice workflows with **AWS Lambda**, writing generated PDFs to **S3** and invoice metadata to **DynamoDB**.
- Crafted financial dashboards in React using **SWR and Chart.js** to visualize revenue trends from **DynamoDB**, with API performance validated via **Postman**.
- Leveraged **Redis Pub/Sub** to broadcast order-status events in real time **across React components**, allowing users to track requests without manual refreshes.
- Architected a **context-aware** notification service using **Socket.io, Node.js, and Redis Pub/Sub** to alert teams of high-value orders and test completions.
- Enabled **targeted Redis-based** push notifications for discount offers and order updates, driving a 20% uplift in repeat test orders.
- Modeled ten **DynamoDB tables** to handle **heterogenous data**—images, links, text, numeric results, and timestamps—ensuring flexible schema and fast reads.
- Built a **customer-management module** in Next.js with DynamoDB backend and **secure session handling**, enabling dynamic discount rules per client.
- Enforced **XSS protection and CSRF prevention** in **Next.js middleware** and **API Gateway endpoints**, hardening the app against web-based attacks.
- Centralized performance logging with **Amazon CloudWatch**, surfacing component render times, API latencies, and **custom business KPIs**.
- Combined **D3.js with Deck.GL** for animated, layered visualizations of delivery routes and geospatial datasets, giving logistics teams deeper insights into real-time supply chain performance.
- Implemented **custom D3.js charting components** for revenue analysis and material testing KPIs, offering clients dynamic filtering, sorting, and drill-down capabilities within the reporting portal.
- Instrumented **Web Vitals** observers on the client to capture **LCP, FID, and CLS**, feeding metrics back into **CloudWatch** for continuous optimization.

Client: Cognizant, India

Role: Full Stack Developer | Aug 2018 – Dec 2019.

Migration of a legacy **Claims Management** system to a **microservices architecture** using **Spring Boot** and **API Gateway**, improving scalability and performance. Integrated **prescription management** and **telehealth** features into the **Healthcare portal**.

- Developed and consumed **SOAP** and **RESTful APIs** with **Spring MVC** and **Spring Web Services** to integrate with legacy mainframe systems and partner portals
- Designed data models with **Hibernate ORM** and **JPA** for core insurance modules (policies, billing, claims), optimizing queries and indexing for performance
- Led migration of a legacy monolithic Claims Management application to a **microservices architecture** using **Spring Cloud** and **API Gateway**, improving system scalability
- Architected and implemented a suite of **Spring Boot microservices** powering the Policy Issuance Engine and Claims Processing Dashboard in the member portal.
- Collaborated with UX designers to bind **Amplify Theme** tokens into React components, ensuring consistent light/dark mode theming and design-to-code fidelity.
- Implemented real-time prescription management features with RxJS, paving the way for **AI-driven medication reminders** and chat-based prescription refills.
- Leveraged **RxJS** for handling asynchronous API calls, ensuring real-time updates of prescription data with zero latency.
- Implemented **JWT** authentication in **Spring Security** to ensure secure and compliant handling of sensitive prescription data, meeting **HIPAA** standards.
- Architected state management with **NgRx Store**, **Actions**, and **Effects** for the Claims Dashboard, enabling predictable state transitions and time-travel debugging
- Integrated **RESTful APIs** via **Angular HttpClient**, implementing global HTTP interceptors for error handling, token refresh, and retry logic to improve resilience under network failures
- Implemented reactive **Angular forms** for policy application workflows, using **custom validators** and dynamic form arrays to provide real-time validation feedback
- Configured client-side routing with **Angular Router** and used route guards to manage authorization.
- Used **RxJS** operators like **switchMap**, **concatMap** and **mergeMap** to manage asynchronous streams for live policy searches and auto-save drafts.
- Improved SEO and performance using **Angular Universal** for server-side rendering of Policy Summary pages.
- Accessibility and UX with ARIA and WCAG standards — foundational for **AI voice-assist and multimodal conversational agents** in healthcare portals.
- Optimized database performance with advanced **SQL** tuning, table partitioning, and materialized views
- Designed and implemented **Spring Batch** jobs for nightly ETL, processing many of policy records into an **Oracle PL/SQL** data warehouse
- Authored comprehensive test suites **JUnit**, **Mockito**, and **Cucumber** BDD scenarios to drive high code coverage and catch integration defects early
- Instrumented logging and monitoring with **Splunk** and **Kibana**, configuring alerts on key SLIs (error rate, response time)
- Streamlined API development using **Swagger/OpenAPI**, generating server stubs and client SDKs for third-party integrations
- Provisioned cloud infrastructure with **Terraform** across **AWS** and **Azure**, managing **VPCs**, **subnets**, and **Kubernetes** clusters (**EKS/AKS**)

Education

- Master's in computer science from Florida Atlantic University - April 2024
- Bachelor's in computer science from Mahatma Gandhi Institute of Technology - April 2018